

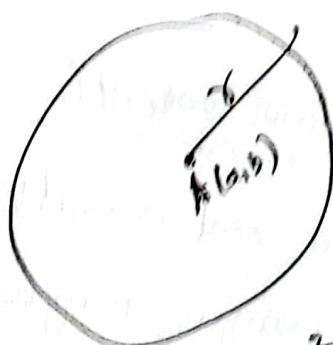
Function of n variables

A function f of n variable real variables

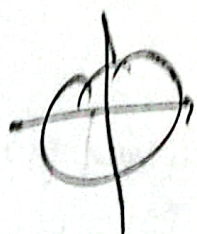
$x_1, x_2, x_3, \dots, x_n$, is a rule that assigns a unique real number.

$w = f(x_1, x_2, x_3, \dots, x_n)$ in some set D of n dimensional space.

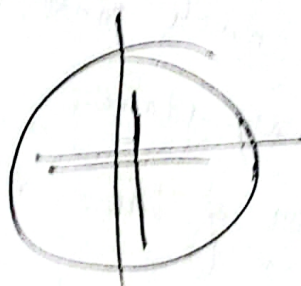
Analytical Geometry



$$(x-a)^2 + (y-b)^2 = r^2$$



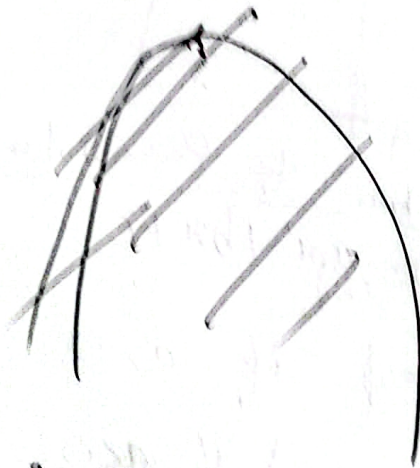
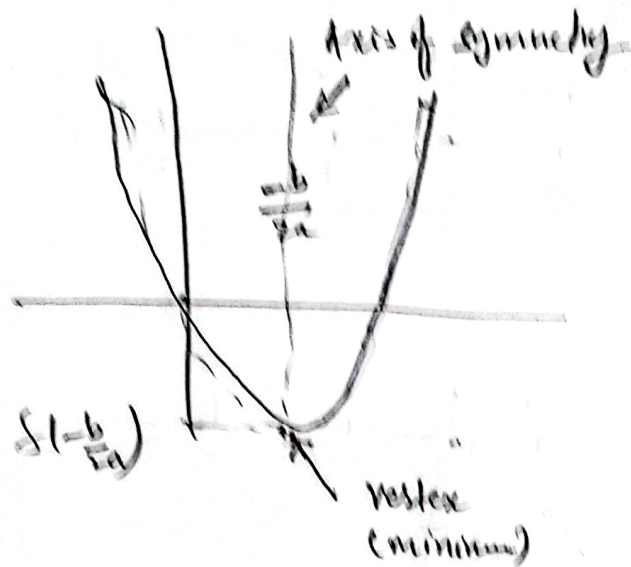
$$x^2 + y^2 = a^2$$



$$x^2 + y^2 \leq a^2$$

$$y = ax^2 + bx + c$$

$$(a > 0)$$



Ex: $y = -x^2 + 4x$

since $a = -1 < 0$, parabola is opening downward.
 vertex occurs at $x = -b/2a = (-4)/2(-1) = 2$

Axis of symmetry is the vertical line $x = 2$.
 the y coordinate of the vertex is

$$y = -(2)^2 + 4(2) = 4$$

Hence vertex is $V(2, 4)$

$$y = -x^2 + 4x$$

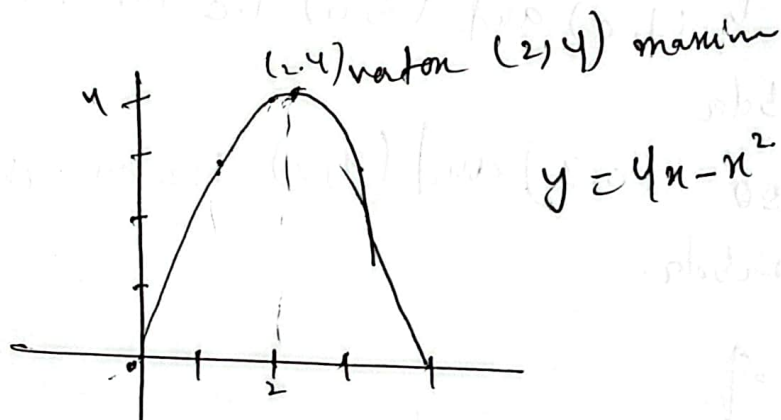
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The zeros of the parabola (i.e. the point where the parabola meets x -axis) are the solutions to

$$-x^2 + 4x = 0 \text{ so } x = 0 \text{ and } x = 4$$

therefore $(0,0)$ and $(4,0)$ lie on the parabola

Also $(1,3)$ and $(3,3)$ lie on the parabola.



Example: $y = x^2 - 4x + 3$

since $a = 1 > 0$, parabola is opening upward

vertex occurs at $x = -b/2a = -(-4)/2(1)$

$$\Rightarrow \frac{4}{2} = 2$$

Axis of symmetry is the vertical line $x = 2$

the y -co-ordinate of the vertex is

$$y = (2)^2 - 4(2) + 3 = -1$$

Hence vertex is $V(-2, -1)$

continued

$$y = x^2 - 4x + 3$$

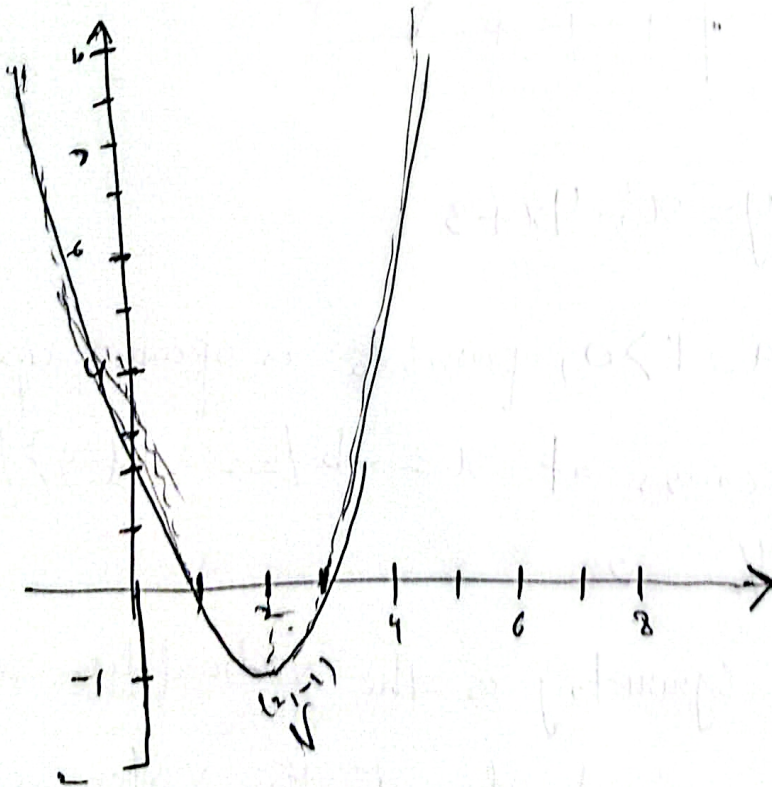
the zeros of the parabola (i.e. the point where the parabola meets x -axis) are the solutions to

$$x^2 - 4x + 3 = 0 \text{ so } x = 1 \text{ \& } x = 3.$$

therefore $(1, 0)$ and $(3, 0)$ lie on the

parabola

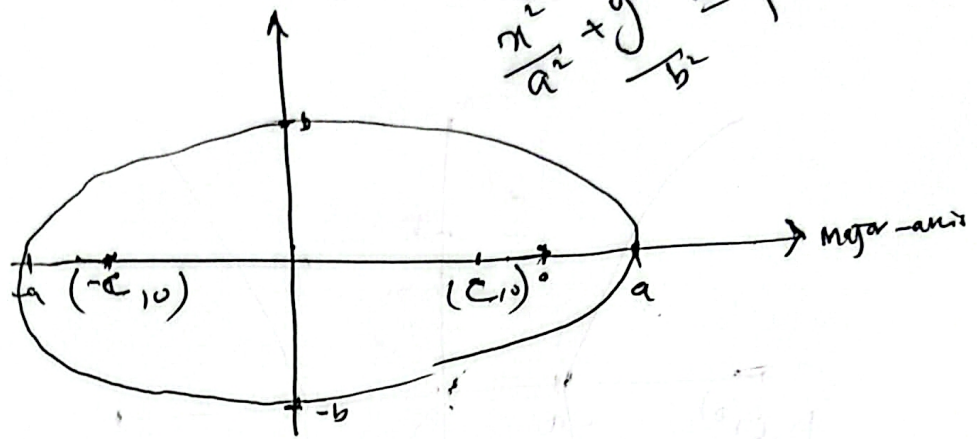
Also $(0, 3)$ and $(4, 3)$ lie on the parabola.



Ellipse

(11)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

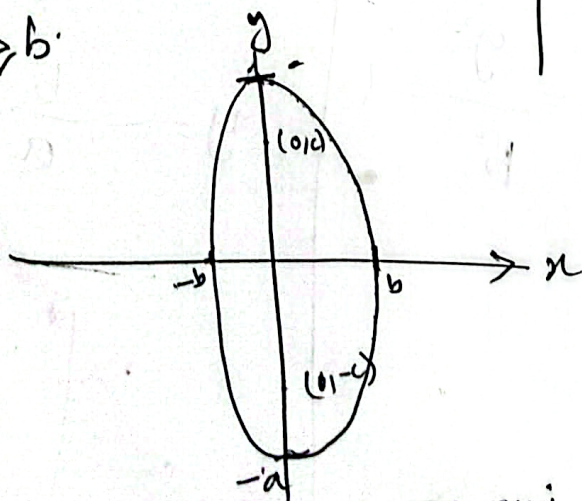


Description

- Foci and major axis on the x-axis
- minor axis is on the y-axis
- centre at origin
- x-intercepts: $\pm a$
- y-intercepts: $\pm b$
- $a > b$

Standard Equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

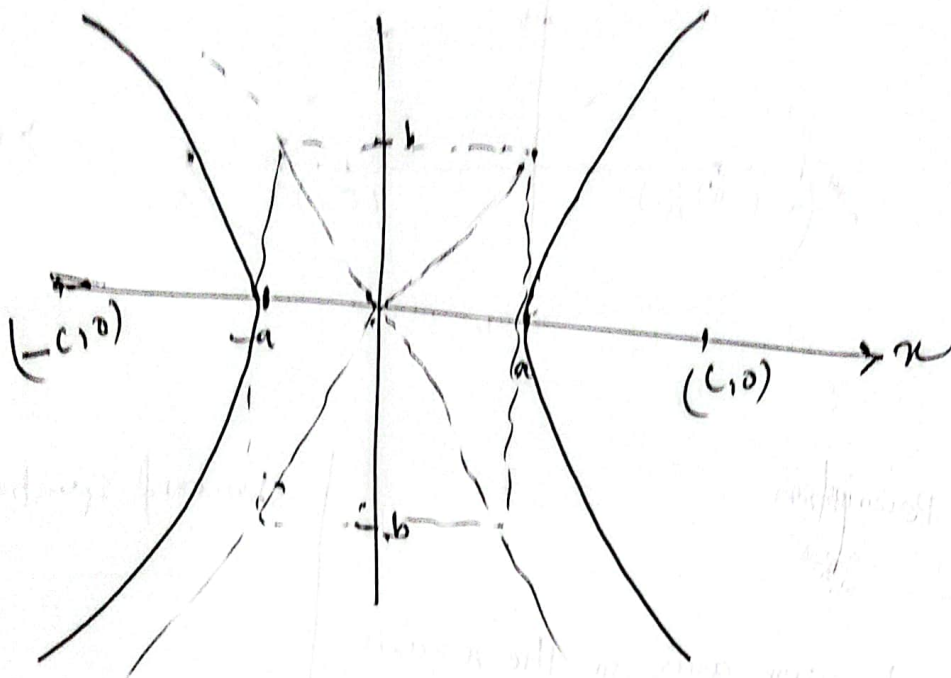


$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

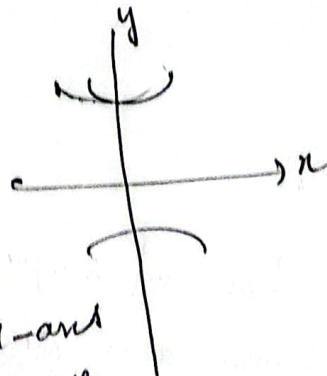
- Foci and major axis on y-axis
- minor axis on x-axis
- centre at origin
- x-intercepts: $\pm b$
- y-intercepts $\pm a$
- $a > b$

Hyperbola

(12)



Description	$\frac{x^2}{a^2}$ standard equation	Asymptote equations
Foci on the x-axis	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$y = \frac{b}{a} x$
Conjugate axis on the y-axis		$y = -\frac{b}{a} x$
Centre at origin		



Foci y-axis
Conjugate x-axis
Centre origin